Engineering a Low-cost, Reusable Surgical Box Model of the Liver for Resident Training

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Background

- Residents benefit from a reliable training method to familiarize themselves with liver transplant procedures before entering the OR:
  - Liver transplant surgery consists of four major stages: donor hepatectomy, recipient hepatectomy, graft implantation, and biliary tract reconstruction.[1]
  - At UF, most general surgery residents practice transplant surgery for 2 of 40 clinical rotations.[2]
- Current training methods present limitations:
  - *Simulations/Videos*: Convenient to set up and distribute, but lack or require integration of haptic feedback.
  - *Cadavers*: Anatomically accurate, but limited by reusability and availability.
- We have constructed a reusable, low-cost 3D box model that allows residents to practice a bicaval liver transplant procedure with anatomically accurate organs and vessels.

Methods

A. **Model Housing:** A custom-designed box features 2 chambers, each with 4 quadrants. Parts were designed in SolidWorks and 3D printed with polylactic acid (PLA) for rigidity, then assembled with epoxy resin and coated with Flex Seal to prevent water damage.
- **Lower Chamber:** Acts as a water reservoir, featuring a sloped bottom giving way to a drain channel that can be plugged during runs.
- **Upper Chamber:** Houses most model components, and is sectioned into three compartments:
  - The **main compartment** contains the liver & its vasculature.
  - The **inferior compartment** houses various fluid uptake ports and a channel to accommodate the wiring for submersible motors.
  - The **lateral compartment** includes the 12V DC wall-connecting power supply.

B. **Fluid Flow System:** Mimics the natural vasculature of the liver, including hepatic artery (HA), portal artery (PA), bile duct (BD), and inferior vena cava (IVC).
- **Tubing:** While **plastic tubing** was used to ensure rigidity in pipes pulling water up from the reservoir, **Penrose tubing** is used in the main model to facilitate suturing of the liver's associated vessels.
- **Motors:** A combination of **submersible and peristaltic motors** housed in the lower chamber and lateral compartment drive fluid flow.
- **Connectors:** Penrose and plastic tubing sections are conjoined with **hose barb fittings** and secured with **hose clamps** to prevent potential separation caused by high-pressure fluid flow.

C. **Organ Models:** The model replicates the liver and duodenum.
- **Liver:** Model sourced from GrabCAD, an open-source library. Modifications include adding ports for fluid vessels and scaling the model up to anatomically accurate dimensions.
- **Duodenum:** Model sourced from Thingiverse, an open-source library. Duodenum model was inverted to create a mold, which was 3D printed as 2 parts. Mold was cast with Flex Foam (urethane mixture) to form a flexible, durable part.

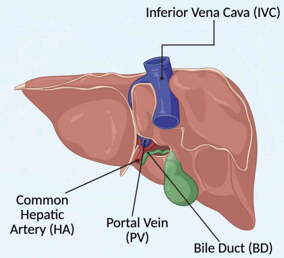

Figure 1: Human liver, posterior view; vasculature indicated

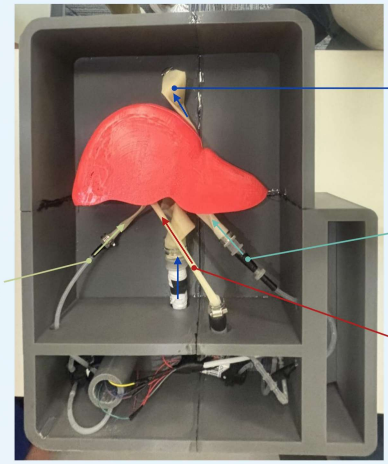

Figure 2: Upper chamber of box model, superior view; vessels and fluid flow directions indicated

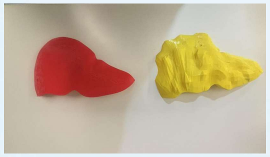

Figure 3: 3D printed models: healthy liver *(left)*; liver under cirrhosis *(right)*

Figure 4: Posterior half of liver model with molding for vasculature

Figure 5: Upper and lower chambers of box model, lateral view

Results

- We have successfully designed a low-cost, 3-D liver transplant model that replicates the vasculature and fluid flow involved in a bicaval liver transplant:
  - Professional students can practice anastomoses of the inferior vena cava, hepatic artery, portal vein, and bile duct with Penrose or cadaver-sourced vessels.
  - Fluid flow through the vessels is functional, and the vessel system remains stable under the forces of flowing liquid.
  - The model costs about $100 to produce.
  - Operating costs are restricted to replacement of fluids and vessels between runs.
- Time to build model: 12-15 hours to design, 8-10 hours to construct

Conclusions

- Our liver transplant model successfully achieves its intended purpose of modeling liver vasculature for surgery while maintaining low production and operation costs.
- This model demonstrates strong potential as a practical tool for surgical training and preoperative planning.

Future Work

- Our next steps are primarily study implementation of the model in a clinical setting:
  - **Model Validation:** Conduct a validation study by recruiting a cohort of liver transplant surgeons to evaluate the model using the MISSES scale, assessing anatomical accuracy and surgical realism.
  - **Educational Effectiveness:** Implement a pretest–posttest study with resident surgeons to evaluate improvements in knowledge and technical skills related to vascular anastomosis in liver transplantation.
- We also intend to explore **patient-specific models** to enhance surgical planning and procedural accuracy.

References

[1]Makowka, L., Stieber, A. C., Sher, L., Kahn, D., Mieles, L., Bowman, J., Marsh, J. W., & Starzl, T. E. (1988). Surgical Technique of Orthotopic Liver Transplantation. Gastroenterology Clinics of North America, 17(1), 33–51

[2]Clinical Training Overview » Department of Surgery » College of Medicine » University of Florida. (n.d.). surgery.med.ufl.edu/education/acute-care-surgery/applicants/yearly-training-overviews/

Acknowledgments